

GUJARAT TECHNOLOGICAL UNIVERSITY

Post Graduate Diploma in Data Science (PGDDS) | Mini Project (DS02080041) | Institute Code: 137

Academic Year 2025-26, Semester-2 | Internal Guide: Komal Prajapati

Domain: FinTech | Digital Payments | Transaction Analytics

Subject Code: DS02080041

UPI Transaction Failure Root-Cause Intelligence System

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PROBLEM STATEMENT

UPI processes billions of transactions daily in India. Transaction failures cause user frustration and direct financial losses for banks and payment platforms.

GOAL: Automatically classify the root cause of each failed UPI transaction into one of 8 failure categories using supervised ML over logs and system indicators.

Key Challenge: Severe class imbalance - Fraud_Flag is only ~2% vs Bank_Server_Down at 38%. SMOTE is used to balance minority classes before model training.

ROOT-CAUSE CLASSES (8 Categories)

- * Bank_Server_Down - server degraded/down, high RT
- * Insufficient_Funds - amount exceeds account balance
- * Wrong_Credentials - incorrect PIN, high retry count
- * Network_Timeout - response time exceeds 5000 ms
- * VPA_Not_Found - payee VPA invalid/unregistered
- * Fraud_Flag - anomalously high amount flagged
- * NPCI_Gateway_Error - NPCI latency spike detected
- * User_Declined - user manually declined payment

DATASET

Records : 100,000 transactions
Failures : ~28,000 (28% failure rate)
Features : 12 input variables
Target : root_cause (8 classes)

Numeric Features:

amount (INR)
response_time_ms
retry_count
hour, day_of_week
npci_gateway_latency_ms

Categorical Features:

payer_bank & payee_bank
transaction_type (P2P/P2M)
device_os (Android/iOS/Web)

METHODOLOGY

Step 1: Data Generation
100K synthetic UPI transaction
logs with realistic distributions

Step 2: EDA
Class imbalance analysis
Correlation heatmap & plots

Step 3: Preprocessing
OneHotEncoder (categoricals)
StandardScaler (numerics)
80/20 stratified split

Step 4: SMOTE Oversampling
Balance rare failure classes

Step 5: Train + Deploy
RandomForest (150 estimators)

MODEL RESULTS

~90%

Overall Accuracy

~0.88

Macro F1-Score

150

RF Estimators

~20K

Training Records

TECH STACK

- * Python 3.11 - Core language
- * pandas & NumPy - Data manipulation
- * scikit-learn - ML pipeline & Random Forest
- * imbalanced-learn - SMOTE oversampling
- * matplotlib & seaborn - EDA visualisations
- * Streamlit - Interactive web app
- * Jupyter Notebook - Analysis & reporting
- * GitHub - Version control & hosting

KEY FINDINGS

- * Bank_Server_Down (38%) is the dominant failure class
- * Network_Timeout flagged when response_time > 5000 ms
- * Wrong_Credentials linked to retry_count spikes
- * Fraud_Flag (2%) benefits most from SMOTE balancing
- * NPCI_Gateway_Error tied to gateway latency > 380 ms
- * Random Forest achieves ~90% on all 8 failure classes
- * Peak failures occur between 00:00 and 03:00 hours
- * Android + 3G combination has highest failure rate

ML PIPELINE FLOW

Data Generation

EDA & Analysis

Preprocessing

SMOTE Balance

Random Forest

Streamlit Deploy

LIVE STREAMLIT APPLICATION

<https://kishan2511-upi-transaction-failure-prediction-app-fv8njm.streamlit.app/>

GitHub Repository

https://github.com/kishan2511/upi_transaction_failure_prediction

